

ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT

MATHEMATICS
Higher 2
Paper 1

9740 / 01

18 August 2009

JC 2 PRELIMINARY EXAMINATION

Time allowed: **3 hours**

Additional Materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Index number, Form Class, graphic and/or scientific calculator model/s on the cover page.

Write your Index number and full name on all the work you hand in.

Write in dark blue or black pen on your answer scripts.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in the question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of **6** printed pages.



Anglo-Chinese Junior College

[Turn Over]

ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT
JC2 Preliminary Examination 2009

MATHEMATICS 9740
Higher 2
Paper 1

/ 100

Index No:

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Form Class: _____

Name: _____

Calculator model: _____

Arrange your answers in the same numerical order.

Place this cover sheet on top of them and tie them together with the string provided.

Question no.	Marks
1	
2	
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12	

- 1 Solve the inequality $x - 2 \geq \frac{5}{x}$ exactly. [3]

Hence deduce the solution of $|x| - 2 > \frac{5}{|x|}$. [2]

- 2 It is given that $U_1 = e^2$ and $U_{r+1} - U_r = (e-1)e^{r+1}$ where $r \in \mathbf{Z}^+$.

By considering $\sum_{r=1}^n (U_{r+1} - U_r)$, show that $U_n = e^{n+1}$. [3]

Hence find the product of the terms $U_2, U_3, U_4, \dots, U_n$ in terms of n . [2]

- 3 Family groups from 4 families with the surnames of Lee, Chan, Lim and Tan consist of

- (a) 2 parents, 2 grandparents and 1 child,
- (b) 1 parent, 2 grandparents and 3 children,
- (c) 2 parents, 1 grandparent and 4 children,
- (d) 1 parent, 1 grandparent and 2 children,

respectively.

All the families book with the same travel company to fly one way from Singapore to Japan. Whilst the parents have to pay the full fare there is a reduced price for the grandparents and an even cheaper price for children. The total cost for the Lee, Chan and Lim families are \$1959, \$2196 and \$2640 respectively. What is the cost for the Tan family? [4]

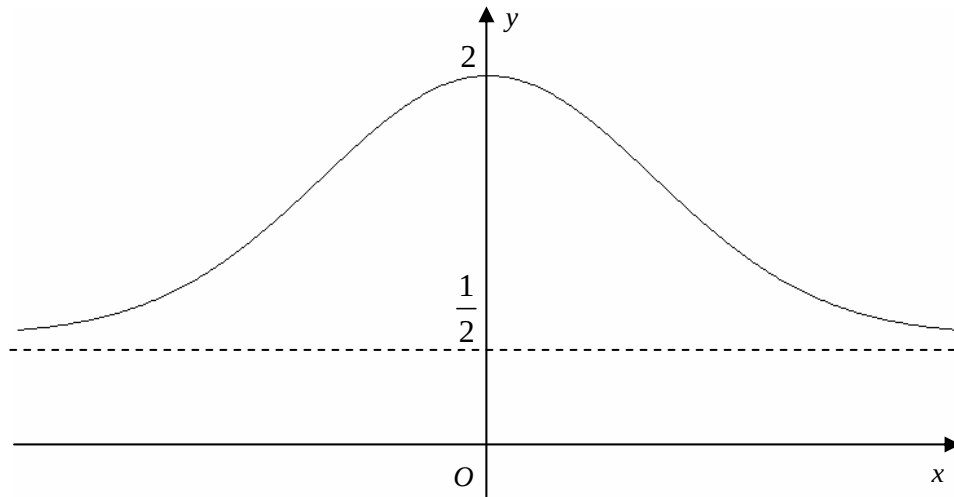
- 4 (i) Evaluate S_1, S_2, S_3 , and S_4 for the sum $S_n = \sum_{r=1}^n \frac{1}{r(r+1)}$. [1]

- (ii) By expressing the values of S_n in the form $[1 - f(n)]$ for $n = 1, 2, 3$ and 4 , obtain a conjecture for $f(n)$ in terms of n . [2]

- (iii) Hence prove by induction the result for S_n for all positive integers n . [4]

[Turn Over]

5



The diagram shows a sketch of the graph of $y = f(x)$ which is symmetrical about the y -axis. The graph of $y = f(x)$ has a maximum point at $(0, 2)$ and a horizontal asymptote at $y = \frac{1}{2}$. On separate diagrams sketch the graphs of

(i) $y = f'(x)$, where $f'(x) = \frac{d}{dx}[f(x)]$, [2]

(ii) $y = \frac{1}{f(x)}$, [2]

(iii) $y^2 = f(x)$, [3]

showing clearly the equations of any asymptote(s), the coordinates of any turning point(s) and axial intercept(s).

6 The functions f and g are defined as follows:

$$f : x \mapsto x^2 - 2x + 2, \quad 0 \leq x \leq 1$$

$$g : x \mapsto \frac{x+2}{x+1}, \quad 0 \leq x \leq 3$$

(i) Show that the inverse function of f exists, and define the inverse function f^{-1} . [4]

(ii) Show that the composite function gf exists, and define it. [3]

(iii) Find the range of gf . [2]

7 A point (x, y) lies on the curve with equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{a}$, where a is a non-zero constant, $x \neq 0$ and $y \neq 0$.

- (i) Express $\frac{dy}{dx}$ in terms of x and y , and explain clearly whether y is an increasing or a decreasing function. [3]
- (ii) State, giving a reason, whether there are any stationary points on this curve. [2]
- (iii) Find the equation of the tangent to the curve at the point $(2a, 2a)$ and determine if the tangent cuts the curve again. [5]

8 (a) Sketch on the same diagram the graphs of $y = \cos^{-1} x$ and $y = \tan^{-1} x$, stating the domain and the range of each function. [4]

(b) Sketch the graph of $y = |\sin^{-1} x|$ and find the exact solutions of $|\sin^{-1} x| = \frac{\pi}{3}$. [3]

(c) State how the graph of $y = e^{-x}$ can be obtained from the graph of $y = -\ln x$. Sketch, on the same axes, the graphs of $y = -\ln x$ and $y = e^{-x}$, showing clearly the geometrical relationship. [3]

Given that α is the root of the equation $e^{-x} = x$, solve the inequality $-\ln x > e^{-x}$, giving your answer in terms of α . [1]

9 The variables x and y are connected by the differential equation

$$\frac{dy}{dx} - 1 - (x + y)^2 = 0.$$

(i) Show that the substitution $u = x + y$ reduces the differential equation to $\frac{du}{dx} - 2 - u^2 = 0$. [2]

(ii) Hence, solve $\frac{dy}{dx} - 1 - (x + y)^2 = 0$, expressing y in terms of x , given that $y = 2$ when $x = 0$. [4]

(iii) The variable A is defined to be the product of x and y , where both x and y are functions of z . If $\frac{dA}{dz} = 3$ when $x = 0$, find the value of $\frac{dx}{dz}$ when $x = 0$. [3]

[Turn Over]

- 10(a)** Without the use of a calculator, find the complex numbers z and w in the form $a + ib$ that satisfy the two simultaneous equations

$$z + (2 + i)w = -9 + 16i \quad \text{and} \quad z^* + w = 3i. \quad [4]$$

- (b)** [Give your answers in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.]
Solve

$$z^3 = i. \quad [3]$$

Deduce the roots of the equation

$$w^3 = -i. \quad [2]$$

- 11(i)** Given that $y = \cot\left(2x + \frac{\pi}{4}\right)$, find an equation relating $\frac{dy}{dx}$ and y . [2]

Hence show that

$$\frac{d^3y}{dx^3} = k \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right],$$

where k is a constant to be found. [2]

- (ii)** Find the Maclaurin's series for y up to and including the term in x^3 if x is sufficiently small for powers of x higher than x^3 to be neglected. [2]

- (iii)** Using the Maclaurin's expansion for y , estimate the value of $\operatorname{cosec}^2\left(\frac{13\pi}{50}\right)$, giving your answer in the form $a + b\pi + c\pi^2$, where a , b and c are constants to be determined. [4]

- 12** The coordinates of the points A and B are $(-1, 2, 4)$ and $(0, -2, 5)$ respectively. The equations of two planes p_1 and p_2 are $x + y + z = 7$ and $x - y + az = 1$ respectively, where a is a constant.

- (i)** Find the position vector of the foot of perpendicular from the point A to the plane p_1 . [4]
- (ii)** The two planes p_1 and p_2 intersect at the line ℓ . Verify that the point with coordinates $(4, 3, 0)$ is a common point on the planes p_1 and p_2 and hence find the equation of the line ℓ . [3]
- (iii)** Another plane p_3 has equation $x + 2y + 3z = b$. Find the values of a and b if all the three planes p_1 , p_2 and p_3 intersect at the line ℓ . [4]
- (iv)** If the length of projection of the line segment AB on the line ℓ is $\frac{1}{\sqrt{6}}$ unit, find the values of a . [3]